

Propositional Logic in Artificial Intelligence

◆ What is it?



- A type of logic that uses **statements (propositions)** to show whether something is **true or false**
- Each statement must be **either true or false** — no in-between
- Used in AI for **reasoning, decision-making, and knowledge representation**

Types of Propositions

1. Atomic Proposition

→ Simple statement, must be true or false

Examples:

- “ $2 + 2 = 4$ ” → True
- “The sun is cold” → False

2. Compound Proposition

→ Combines atomic propositions using logical symbols

Examples:

- “It is raining AND the street is wet”
- “Ankit is a doctor AND his clinic is in Mumbai”

Logical Connectives (with Easy Examples)

Connective	Symbol	Meaning	Example Sentence	Logic Form
Negation	$\neg P$	Not	“It is NOT raining”	$\neg P$
Conjunction	$P \wedge Q$	AND	“Rohan is intelligent AND hardworking”	$P \wedge Q$
Disjunction	$P \vee Q$	OR	“Ritika is a doctor OR engineer”	$P \vee Q$
Implication	$P \rightarrow Q$	If...then	“If it rains, then the street is wet”	$P \rightarrow Q$
Biconditional	$P \Leftrightarrow Q$	If and only if (both true)	“If I’m breathing, then I’m alive”	$P \Leftrightarrow Q$

Connective symbols	Word	Technical Term	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\leftrightarrow$	If and only if	Biconditional	$A \leftrightarrow B$
$\neg$ or $\sim$	Not	Negation	$\neg A$ or $\neg B$

Truth Table

In propositional logic, we need to know the truth values of propositions in all possible scenarios. We can combine all the possible combinations with logical connectives, and the representation of these combinations in a tabular format is called a Truth table. Following is the truth table for all logical connectives:

For Negation :

P	$\neg P$
True	False
False	True

For Conjunction:

P	Q	$P \wedge Q$
True	True	True
True	False	False
False	True	False
False	False	False

For disjunction:

P	Q	$P \vee Q$
True	True	True
False	True	True
True	False	True
False	False	False

For Implication:

P	Q	$P \rightarrow Q$
True	True	True
True	False	False
False	True	True
False	False	True

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For Biconditional :

P	Q	$P \leftrightarrow Q$
True	True	True
True	False	False
False	True	False
False	False	True

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Truth Table with Three Propositions

We can build a proposition composing three propositions: P, Q, and R. This truth table is made up of 8n Tuples as we have taken three proposition symbols.

P	Q	R	$\neg R$	$P \vee Q$	$P \vee Q \rightarrow \neg R$
True	True	True	True	True	True
True	True	True	True	True	True
True	True	True	True	True	True
True	True	True	True	True	True
False	False	False	False	False	False
False	False	False	False	False	False
False	False	False	False	False	False
False	False	False	False	False	False

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Precedence of connectives:

Just like arithmetic operators, there is a precedence order for propositional connectors or logical operators. This order should be followed while evaluating a propositional problem. Following is the list of the precedence order for operators:

 Order	 Operator	 Description	 Example
1st	Parenthesis ()	Grouping expressions	$(P \wedge Q) \vee R$
2nd	Negation ($\neg$)	NOT	$\neg P$
3rd	Conjunction ($\wedge$)	AND	$P \wedge Q$
4th	Disjunction ($\vee$)	OR	$P \vee Q$
5th	Implication ($\rightarrow$)	IF...THEN	$P \rightarrow Q$
6th	Biconditional ($\Leftrightarrow$)	IF AND ONLY IF	$P \Leftrightarrow Q$

Logical Equivalence

Propositional logic is one of the features that have logical equivalence. The definition of logical equivalence is given by saying two propositions are logically equivalent if and only if the columns of a truth table are the same.

Assuming two propositions, A and B, we'll mark it by $A \Leftrightarrow B$, and this is the logical equivalence. From the below truth table, we see that the columns of $\neg A \vee B$ and $B \rightarrow A$ are identical, so A is Equivalent to B.

A	B	$\neg A$	$\neg A \vee B$	$A \rightarrow B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Properties of Operators

মূল সূত্র ও নিয়ম

- **Commutativity:** $P \wedge Q = Q \wedge P, P \vee Q = Q \vee P$
- **Associativity:** $(P \wedge Q) \wedge R = P \wedge (Q \wedge R)$
- **Identity:** $P \wedge \text{True} = P, P \vee \text{False} = P$
- **Distributive Law:** $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$
- **De Morgan's Law:** $\neg(P \wedge Q) = (\neg P) \vee (\neg Q)$

Logical Laws in Propositional Logic

1. Commutativity

→ Order doesn't matter

- $P \wedge Q = Q \wedge P$
- $P \vee Q = Q \vee P$

Example:

"Rita and Sam are friends" = "Sam and Rita are friends"

2. Associativity

→ Grouping doesn't change result

- $(P \wedge Q) \wedge R = P \wedge (Q \wedge R)$
- $(P \vee Q) \vee R = P \vee (Q \vee R)$

Example:

"(A and B) and C" = "A and (B and C)"

3. Identity Element

→ Combining with True keeps the original

- $P \wedge \text{True} = P$
- $P \vee \text{True} = \text{True}$

Example:

“Rain AND True” = “Rain”

“Rain OR True” = “True”

4. Distributive Law

→ Like math, distributes over AND/OR

- $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$

Example:

“Study AND (Math OR Science)” = “(Study AND Math) OR (Study AND Science)”

5. De Morgan's Laws

→ Negation flips AND/OR

- $\neg(P \wedge Q) = \neg P \vee \neg Q$
- $\neg(P \vee Q) = \neg P \wedge \neg Q$

Example:

“NOT (Rain AND Cold)” = “NOT Rain OR NOT Cold”

6. Double-Negation Elimination

→ Two NOTs cancel out

- $\neg(\neg P) = P$

Example:

“NOT (NOT Happy)” = “Happy”

Uses of Propositional Logic in AI

1. Storing Knowledge (Knowledge Representation)

AI uses propositional logic to save facts in a clear and organized way. This helps the system understand and use information properly.

Example:

“If it rains $\rightarrow$ the road gets wet”

2.Solving Problems & Planning

AI uses logic to figure out what steps to take to reach a goal.

Example:

“To enter a room $\rightarrow$ first open the door”

3.Making Decisions

AI uses logical rules to choose the best option from many choices.

Example:

“If battery is low $\rightarrow$ charge the device”

4.Understanding Language (Natural Language Processing – NLP)

AI turns human sentences into logical form to understand their meaning.

Example:

“If it rains $\rightarrow$ the street gets wet”

This helps AI make smart conclusions from what people say.

Important Facts About Propositional Logic

- It is also called **Boolean Logic** because it works with **True (1)** and **False (0)**
 - Propositions are written using letters like **A, B, P, Q**
 - A proposition is **either True or False**, not both
 - It uses **objects, relations, and logical symbols** to connect ideas
 - **Logical connectives** (like AND, OR, NOT) join statements together
 - If a statement is **always true**, it's called a **Tautology**
 - If a statement is **always false**, it's called a **Contradiction**
 - **Questions or opinions** (like “How are you?”) are **not** propositions
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Limitations of Propositional Logic

- Can't express general ideas like “All humans are mortal”
 - If there are many statements, the **truth table gets too big**
 - Only works with **True/False**, not with **maybe** or **uncertain** info
 - Can't use special symbols like $\forall$ (**for all**) or $\exists$ (**there exists**)
 - Doesn't understand **context or meaning** in complex situations
 - Can't handle **uncertainty or incomplete data**
 - Can't express ideas like **ALL, SOME, or NONE**
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Rules of Inference in Artificial intelligence

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What is Inference?

Inference means **drawing new conclusions** from known facts or evidence. In AI, it helps computers **think logically** and **make decisions** based on what they already know.

What are Inference Rules?

Inference rules are like **formulas or patterns** that help AI create valid arguments. They are used to build **proofs** — step-by-step logic that leads to a final answer or goal.

In inference rules, the implication among all the connectives plays an important role. Following are some terminologies related to inference rules:

- **Implication:** It is one of the **logical connectives which can be represented as $P \rightarrow Q$** . It is a Boolean expression.
- **Converse:** The converse of implication, which means **the right-hand side proposition goes to the left-hand side and vice-versa**. It can be written as $Q \rightarrow P$.
- **Contrapositive:** The negation of converse is termed as **contrapositive**, and it can be represented as $\neg Q \rightarrow \neg P$.
- **Inverse:** The negation of implication is called inverse. It can be represented as $\neg P \rightarrow \neg Q$.

From the above term some of the compound statements are equivalent to each other, which we can prove using truth table:

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$\neg Q \rightarrow \neg P$	$\neg P \rightarrow \neg Q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

Hence from the above truth table, we can prove that $P \rightarrow Q$ is equivalent to $\neg Q \rightarrow \neg P$, and $Q \rightarrow P$ is equivalent to $\neg P \rightarrow \neg Q$.

Types of Inference rules:

1. Modus Ponens:

যদি P সত্য হয় এবং $P \rightarrow Q$ সত্য হয়, তাহলে আমরা Q-ও সত্য বলে ধরে নিতে পারি।

The Modus Ponens rule is one of the most important rules of inference, and it states that if P and $P \rightarrow Q$ is true, then we can infer that Q will be true. It can be represented as:

$$\text{Notation for Modus ponens: } \frac{P \rightarrow Q, P}{\therefore Q}$$

Example:

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I am sleepy" $\implies P$

Conclusion: "I go to bed." $\implies Q$.

Hence, we can say that, if $P \rightarrow Q$ is true and P is true then Q will be true.

Proof by Truth table:

P	Q	$P \rightarrow Q$
0	0	0
0	1	1
1	0	0
1	1	1

2. Modus Tollens:

Modus Tollens:

যদি $P \rightarrow Q$ এবং Q মিথ্যা হয় ($\neg Q$), তাহলে P -ও মিথ্যা হবে ($\neg P$)।

উদাহরণ: যদি "বৃষ্টি হলে রাস্তা ভিজবে" এবং "রাস্তা ভিজেনি" $\rightarrow$ তাহলে "বৃষ্টি হয়নি"

The Modus Tollens rule state that if $P \rightarrow Q$ is true and $\neg Q$ is true, then $\neg P$ will also true. It can be represented as:

$$\text{Notation for Modus Tollens: } \frac{P \rightarrow Q, \neg Q}{\neg P}$$

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I do not go to the bed." $\implies \neg Q$

Statement-3: Which infers that "I am not sleepy" $\implies \neg P$

Proof by Truth table:

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	0
1	1	0	0	1

3. Hypothetical Syllogism:

Hypothetical Syllogism: যদি $P \implies Q$ এবং $Q \implies R$, তাহলে $P \implies R$ হবে।

উদাহরণ: যদি "বৃষ্টি $\implies$ রাস্তা ভিজবে" এবং "রাস্তা ভিজবে $\implies$ পিচ্ছিল হবে" $\rightarrow$ তাহলে "বৃষ্টি $\implies$ পিচ্ছিল হবে"

The Hypothetical Syllogism rule state that if $P \rightarrow R$ is true whenever $P \rightarrow Q$ is true, and $Q \rightarrow R$ is true. It can be represented as the following notation:

Example:

$$\frac{P \rightarrow Q}{Q \rightarrow R} \\ P \rightarrow R$$

Statement-1: If you have my home key then you can unlock my home. $P \rightarrow Q$

Statement-2: If you can unlock my home then you can take my money. $Q \rightarrow R$

Conclusion: If you have my home key then you can take my money. $P \rightarrow R$

Proof by truth table:

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	1

4. Disjunctive Syllogism:

Disjunctive Syllogism:
 যদি $P \vee Q$ এবং P মিথ্যা হয়, তাহলে Q সত্য হবে।
 উদাহরণ: "চা অথবা কফি", এবং "চা নয়" $\rightarrow$ তাহলে "কফি"

The Disjunctive syllogism rule state that if $P \vee Q$ is true, and $\neg P$ is true, then Q will be true. It can be represented as:

$$\text{Notation of Disjunctive syllogism: } \frac{P \vee Q, \neg P}{Q}$$

Example:

Statement-1: Today is Sunday or Monday. $\implies P \vee Q$

Statement-2: Today is not Sunday. $\implies \neg P$

Conclusion: Today is Monday. $\implies Q$

Proof by truth-table:

P	Q	$\neg P$	$P \vee Q$
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1

5. Addition:

Addition:
 যদি P সত্য হয়, তাহলে $P \vee Q$ সত্য হবে।
 উদাহরণ: "রোদ আছে" $\rightarrow$ "রোদ আছে অথবা মেঘলা"

The Addition rule is one the common inference rule, and it states that If P is true, then $P \vee Q$ will be true.

$$\text{Notation of Addition: } \frac{P}{P \vee Q}$$

Example:

Statement: I have a vanilla ice-cream. $\implies P$

Statement-2: I have Chocolate ice-cream.

Conclusion: I have vanilla or chocolate ice-cream. $\implies (P \vee Q)$

Proof by Truth-Table:

P	Q	$P \vee Q$
0	0	0
1	0	1
0	1	1
1	1	1

6. Simplification:

Simplification:
যদি $P \wedge Q$ সত্য হয়, তাহলে P-ও সত্য হবে।
উদাহরণ: “গরম এবং আর্দ্র” $\rightarrow$ “গরম”

The simplification rule state that if $P \wedge Q$ is true, then Q or P will also be true. It can be represented as:

$$\text{Notation of Simplification rule: } \frac{P \wedge Q}{Q} \text{ Or } \frac{P \wedge Q}{P}$$

Proof by Truth-Table:

P	Q	$P \wedge Q$
0	0	0
1	0	0
0	1	0
1	1	1

7. Resolution:

Resolution:
যদি $P \vee Q$ এবং $\neg Q \vee R$ সত্য হয়, তাহলে $P \vee R$ সত্য হবে।
উদাহরণ: “বৃষ্টি অথবা মেঘলা” এবং “মেঘলা নয় অথবা ঠান্ডা” $\rightarrow$ তাহলে “বৃষ্টি অথবা ঠান্ডা”

The Resolution rule state that if $P \vee Q$ and $\neg P \vee R$ is true, then $Q \vee R$ will also be true. It can be represented as

$$\text{Notation of Resolution } \frac{P \vee Q, \neg P \vee R}{Q \vee R}$$

Proof by Truth-Table:

P	$\neg P$	Q	R	$P \vee Q$	$\neg P \vee R$	$Q \vee R$
0	1	0	0	0	0	0
0	1	0	1	0	0	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	0	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1

The Wumpus World in Artificial intelligence

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Certainly! Here's a side-by-side explanation of **Wumpus World in Artificial Intelligence** in both **English** and **Bangla**, written in point form for easy understanding:

Wumpus World in Artificial Intelligence

◆ Introduction | পরিচিতি

English:

- Wumpus World is a simple artificial world used to show how a knowledge-based agent works.
- It was inspired by the 1973 video game *Hunt the Wumpus*.
- The world is a 4x4 cave with 16 rooms connected by paths.

◆ Main Elements

English:

1. **Wumpus** – A monster that eats the agent if entered. It is static and can be killed with one arrow.
2. **Pits** – Bottomless holes. If the agent falls in, it gets stuck forever.
3. **Gold** – One room may contain gold. The agent must find it and escape.
4. **Agent** – Starts at room (1,1) and tries to find gold without falling or getting eaten.

বাংলা:

1. **Wumpus** – একটি ভয়ংকর দানব, যেটি তার ঘরে ঢুকলে এজেন্টকে খেয়ে ফেলে। এটি স্থির এবং একবার তীর ছুঁড়ে মারা যায়।
2. **Pits** – কিছু ঘরে অন্তহীন গর্ত আছে। এজেন্ট পড়লে চিরতরে আটকে যায়।
3. **Gold** – একটি ঘরে সোনার স্তূপ থাকতে পারে। এজেন্টকে এটি খুঁজে বের করতে হয়।
4. **Agent** – (1,1) ঘর থেকে শুরু করে এবং সোনা খুঁজে বের করে নিরাপদে বেরিয়ে আসার চেষ্টা করে।

◆ Sensory Clues

English:

1. **Stench** – Smell near Wumpus
2. **Breeze** – Wind near pits
3. **Glitter** – Shine in the room with gold
4. **Scream** – Heard if Wumpus is killed

বাংলা:

1. **Stench** – Wumpus-এর পাশের ঘরগুলোতে দুর্গন্ধ থাকে
 2. **Breeze** – গর্তের পাশের ঘরগুলোতে বাতাস অনুভব হয়
 3. **Glitter** – সোনার ঘরে ঝিলমিল দেখা যায়
 4. **Scream** – Wumpus মারা গেলে গুলে গুলার যেকোনো জায়গা থেকে চিৎকার শোনা যায়
-

◆ Goal of the Agent

English:

- Find the gold
- Avoid Wumpus and pits
- Escape safely
- Reward for success, penalty for failure

বাংলা:

- সোনা খুঁজে বের করা
 - Wumpus ও গর্ত এড়ানো
 - নিরাপদে বেরিয়ে আসা
 - সফল হলে পুরস্কার, ব্যর্থ হলে শাস্তি
-

Note: Here Wumpus is static and cannot move.

Following is a sample diagram for representing the Wumpus world. It is showing some rooms with Pits, one room with Wumpus and one agent at (1, 1) square location of the world.

4	 Stench		 Breeze	 PIT
3	 Wumpus	 Breeze  Stench  Gold	 PIT	 Breeze
2	 Stench		 Breeze	
1	 Agent	 Breeze	 PIT	 Breeze
	1	2	3	4

There are also some components which can help the agent to navigate the cave. These components are given as follows:

1. The rooms adjacent to the Wumpus room are smelly, so that it would have some stench.
2. The room adjacent to PITs has a breeze, so if the agent reaches near to PIT, then he will perceive the breeze.
3. There will be glitter in the room if and only if the room has gold.
4. The Wumpus can be killed by the agent if the agent is facing to it, and Wumpus will emit a horrible scream which can be heard anywhere in the cave.

- Wumpus-এর পাশের ঘরগুলোতে দুর্গন্ধ থাকে, তাই এজেন্ট সেই ঘরে গেলে দুর্গন্ধ অনুভব করবে।
- গর্তের (PIT) পাশের ঘরগুলোতে বাতাস থাকে, তাই এজেন্ট যদি গর্তের কাছে যায়, তাহলে সে হালকা বাতাস অনুভব করবে।
- যদি কোনো ঘরে সোনা থাকে, তাহলে সেই ঘরে ঝিলঝিল (glitter) দেখা যাবে — শুধুমাত্র সোনার ঘরেই এটি দেখা যায়।
- এজেন্ট যদি Wumpus-এর দিকে মুখ করে থাকে, তাহলে সে Wumpus-কে হত্যা করতে পারে, এবং Wumpus মারা গেলে একটি ভয়ংকর চিৎকার শোনা যায় যা গুহার যেকোনো জায়গা থেকে শোনা যায়।

PEAS description of Wumpus world:

To explain the Wumpus world we have given PEAS description as below:

Performance measure:

- +1000 reward points if the agent comes out of the cave with the gold.

- -1000 points penalty for being eaten by the Wumpus or falling into the pit.
- -1 for each action, and -10 for using an arrow.
- The game ends if either agent dies or came out of the cave.

Environment:



A 4×4 grid of rooms.
Agent starts at square [1,1], facing right.
Wumpus and gold are randomly placed (not in [1,1]).
Each square (except [1,1]) has a 0.2 chance of containing a pit.

- A 4*4 grid of rooms.
- The agent initially in room square [1, 1], facing toward the right.
- Location of Wumpus and gold are chosen randomly except the first square [1,1].
- Each square of the cave can be a pit with probability 0.2 except the first square.

Actuators (এজেন্টের কার্যকরী অঙ্গ)

এজেন্ট যেসব কাজ করতে পারে:

- **Left turn** – বাম দিকে ঘোরা
- **Right turn** – ডান দিকে ঘোরা
- **Move forward** – সামনে যাওয়া
- **Grab** – জিনিস তোলা (যেমন Gold)
- **Release** – নামানো
- **Shoot** – তীর ছোড়া

Sensors (Wumpus World):
Stench: Agent smells it in rooms next to Wumpus (not diagonally).
Breeze: Felt in rooms next to a Pit.
Glitter: Seen in the room with Gold.
Bump: Felt when agent hits a wall.
Scream: Heard throughout cave when Wumpus is killed.

Sensors:

- The agent will perceive the **stench** if he is in the room adjacent to the Wumpus. (Not diagonally).
- The agent will perceive **breeze** if he is in the room directly adjacent to the Pit.
- The agent will perceive the **glitter** in the room where the gold is present.
- The agent will perceive the **bump** if he walks into a wall.
- When the Wumpus is shot, it emits a horrible **scream** which can be perceived anywhere in the cave.
- These percepts can be represented as five element list, in which we will have different indicators for each sensor.
- Example if agent perceives stench, breeze, but no glitter, no bump, and no scream then it can be represented as:
[Stench, Breeze, None, None, None].

The Wumpus world Properties:

- **Partially observable:** The Wumpus world is **partially observable** because the agent can only perceive the close environment such as an adjacent room.
- **Deterministic:** It is deterministic, as the result and outcome of the world are already known.
- **Sequential:** The order is important, so it is sequential.
- **Static:** It is static as Wumpus and Pits are not moving.
- **Discrete:** The environment is discrete.
- **One agent:** The environment is a single agent as we have one agent only and Wumpus is not considered as an agent.

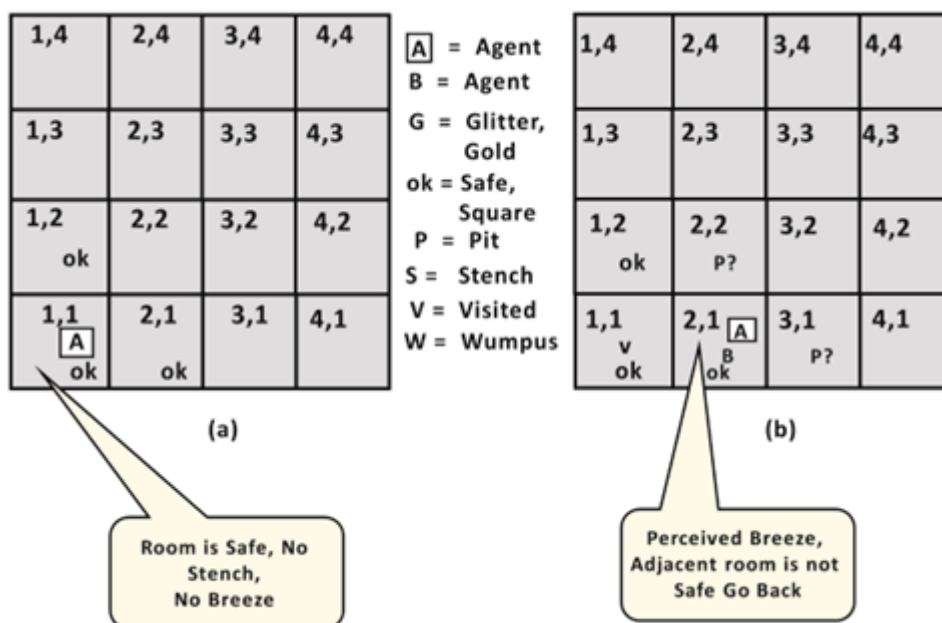
Exploring the Wumpus world:

Now we will explore the Wumpus world and will determine how the agent will find its goal by applying logical reasoning.

Agent's First step:

Initially, the agent is in the first room or on the square [1,1], and we already know that this room is safe for the agent, so to represent on the below diagram (a) that room is safe we will add symbol OK. Symbol A is used to represent agent, symbol B for the breeze, G for Glitter or gold, V for the visited room, P for pits, W for Wumpus.

At Room [1,1] agent does not feel any breeze or any Stench which means the adjacent squares are also OK.



Agent's second Step:

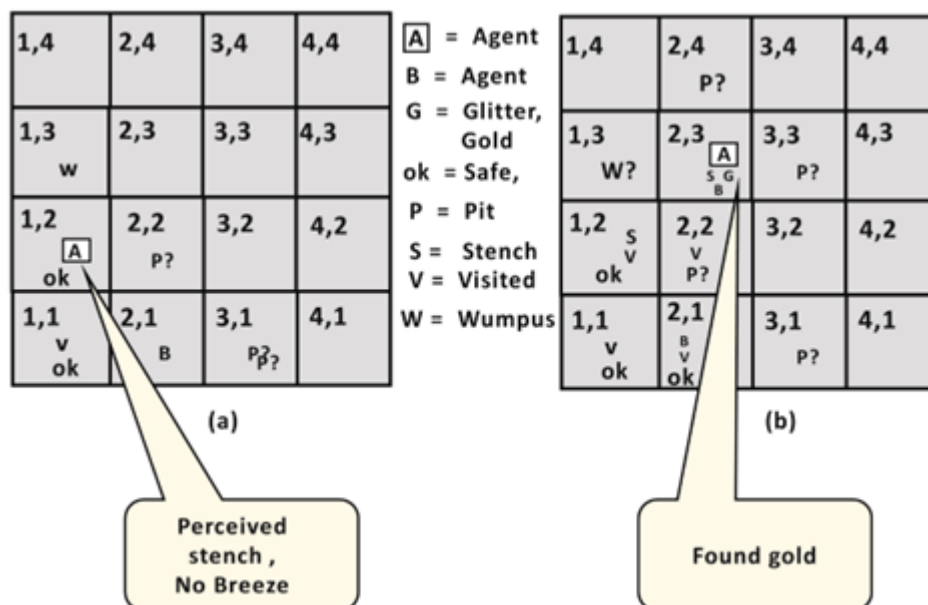
Now agent needs to move forward, so it will either move to [1, 2], or [2,1]. Let's suppose agent moves to the room [2, 1], at this room agent perceives some breeze which means Pit is around this room. The pit can be in [3, 1], or [2,2], so we will add symbol P? to say that, is this Pit room?

Now agent will stop and think and will not make any harmful move. The agent will go back to the [1, 1] room. The room [1,1], and [2,1] are visited by the agent, so we will use symbol V to represent the visited squares.

Agent's third step:

At the third step, now agent will move to the room [1,2] which is OK. In the room [1,2] agent perceives a stench which means there must be a Wumpus nearby. But Wumpus cannot be in the room [1,1] as by rules of the game, and also not in [2,2] (Agent had not detected any

stench when he was at [2,1]). Therefore agent infers that Wumpus is in the room [1,3], and in current state, there is no breeze which means in [2,2] there is no Pit and no Wumpus. So it is safe, and we will mark it OK, and the agent moves further in [2,2].



Agent's fourth step:

At room [2,2], here no stench and no breezes present so let's suppose agent decides to move to [2,3]. At room [2,3] agent perceives glitter, so it should grab the gold and climb out of the cave.

Knowledge Base for the Wumpus World

10 Jun 2025 | 7 min read

As in the previous topic, we have learned about the wumpus world and how a knowledge-based agent evolves the world. Now, in this topic, we will create a knowledge base for the Wumpus world and will derive some proof for the Wumpus world using propositional logic.

The agent starts visiting from the first square [1, 1], and we already know that this room is safe for the agent. To build a knowledge base for the Wumpus world, we will use some rules and atomic propositions. We need a symbol [i, j] for each location in the wumpus world, where i is for the location of rows, and j is for the column location.

1,4	2,4 P?	3,4	4,4
1,3 W?	2,3 S G B	3,3	4,3
1,2	2,2 V P?	3,2	4,2
1,1 A ok	2,1 B V ok	3,1 P?	4,1

Atomic proposition variable for the Wumpus world:

- Let $P_{i,j}$ be true if there is a Pit in the room $[i, j]$.
- Let $B_{i,j}$ be true if the agent perceives a breeze in $[i, j]$ (dead or alive).
- Let $W_{i,j}$ be true if there is a wumpus in the square $[i, j]$.
- Let $S_{i,j}$ be true if the agent perceives a stench in the square $[i, j]$.
- Let $V_{i,j}$ be true if that square $[i, j]$ is visited.
- Let $G_{i,j}$ be true if there is gold (and glitter) in the square $[i, j]$.
- Let $OK_{i,j}$ be true if the room is safe.

*Note: For a 4 * 4 square board, there will be $7*4*4 = 112$ propositional variables.*

Some Propositional Rules for the Wumpus World

$$(R1) \neg S_{11} \Rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

$$(R2) \neg S_{21} \Rightarrow \neg W_{11} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

$$(R3) \neg S_{12} \Rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$$

$$(R4) S_{12} \Rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$$

Note: lack of variables gives us similar rules for each cell.

Representation of Knowledgebase for Wumpus World

Following is the Simple KB for the wumpus world when an agent moves from room [1, 1] to room [2,1]:

$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}
$\neg W_{12}$	----	$\neg P_{12}$	-----	----	$\neg V_{12}$	OK_{12}
$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}

Tpoint Tech

Here in the **first row**, we have mentioned propositional variables for room[1,1], which shows that the room does not have wumpus($\neg W_{11}$), no stench ($\neg S_{11}$), no Pit($\neg P_{11}$), no breeze($\neg B_{11}$), no gold ($\neg G_{11}$), visited (V_{11}), and the room is Safe(OK_{11}).

In the **second row**, we have mentioned propositional variables for room [1,2], which shows that there is no wumpus, stench, and breeze are unknown as an agent has not visited room [1,2], no Pit, not visited yet, and the room is safe.

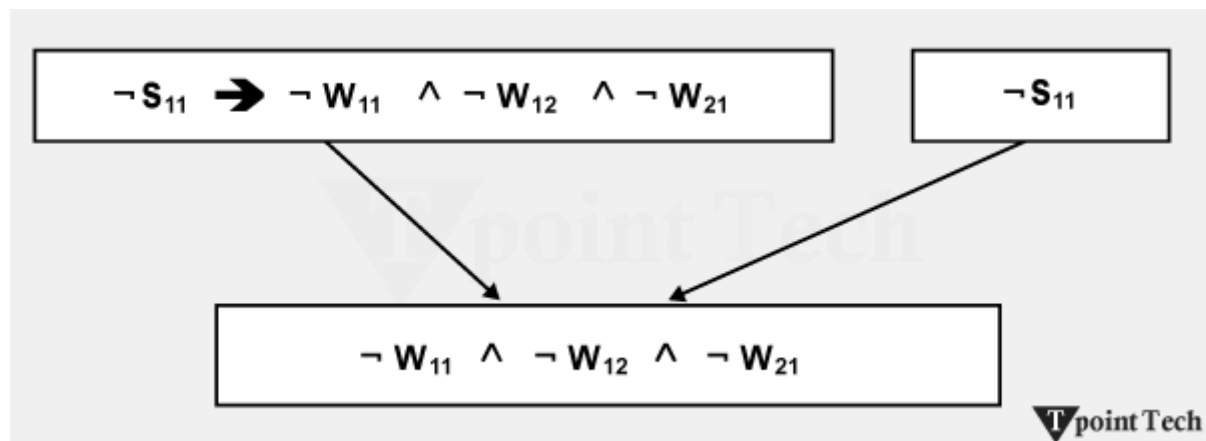
In the **third row**, we have mentioned the propositional variable for room[2,1], which shows that there is no wumpus($\neg W_{21}$), no stench ($\neg S_{21}$), no Pit ($\neg P_{21}$), Perceives breeze(B_{21}), no glitter($\neg G_{21}$), visited (V_{21}), and room is safe (OK_{21}).

Prove that Wumpus is in the room (1, 3)

We can prove that the wumpus is in the room (1, 3) using propositional rules that we have derived for the wumpus world and using the inference rule.

Apply Modus Ponens with $\neg S_{11}$ and R1:

We **will firstly apply MP rule with R1 which** is $\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, and $\neg S_{11}$ which will give the output $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$.



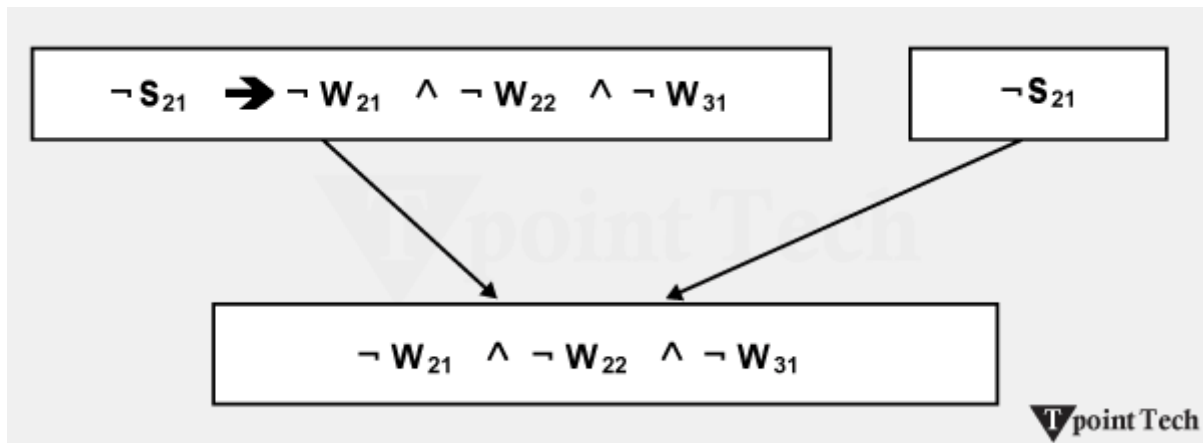
Apply And-Elimination Rule:

After applying And-elimination rule to $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, we will get three statements:

1. $\neg W_{11}$, $\neg W_{12}$, and $\neg W_{21}$

Apply Modus Ponens to $\neg S_{21}$, and R2:

Now we will apply Modus Ponens to $\neg S_{21}$ and R2 which is $\neg S_{21} \rightarrow \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$, which will give the Output as $\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$



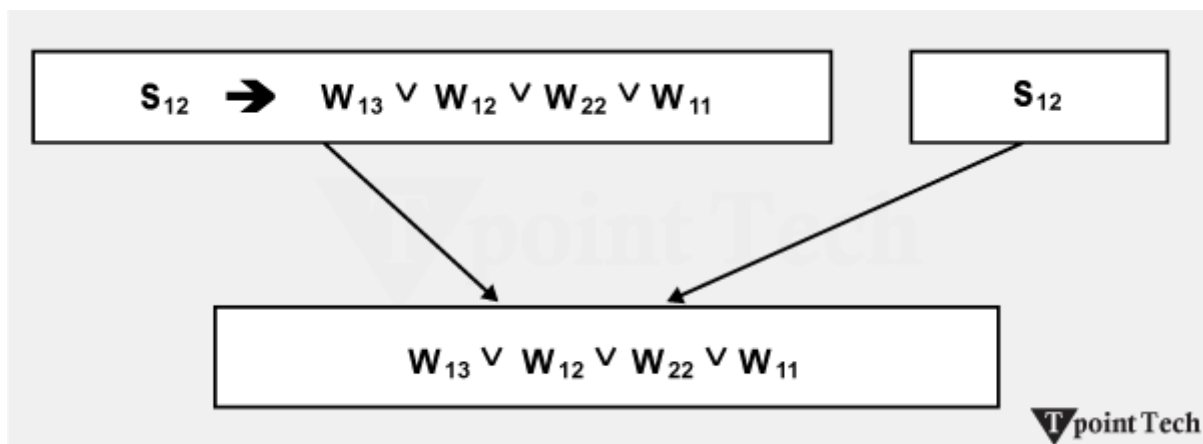
Apply -Elimination Rule:

Now again apply And-elimination rule to $\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$, We will get three statements:

1. $\neg W_{21}$, $\neg W_{22}$, and $\neg W_{31}$

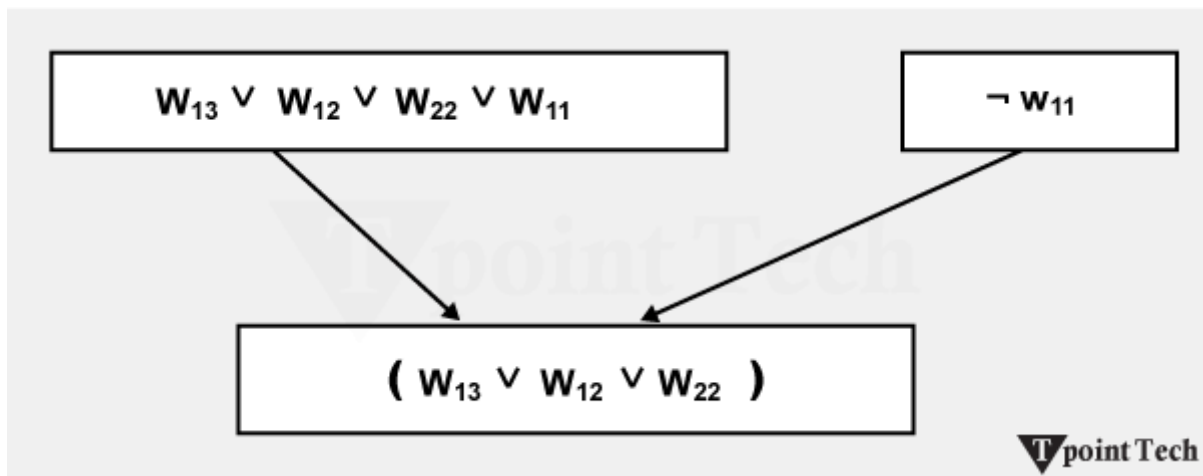
Apply MP to S_{12} and R4:

Apply Modus Ponens to S_{12} and R4, which is $S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$, we will get the output as $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$.



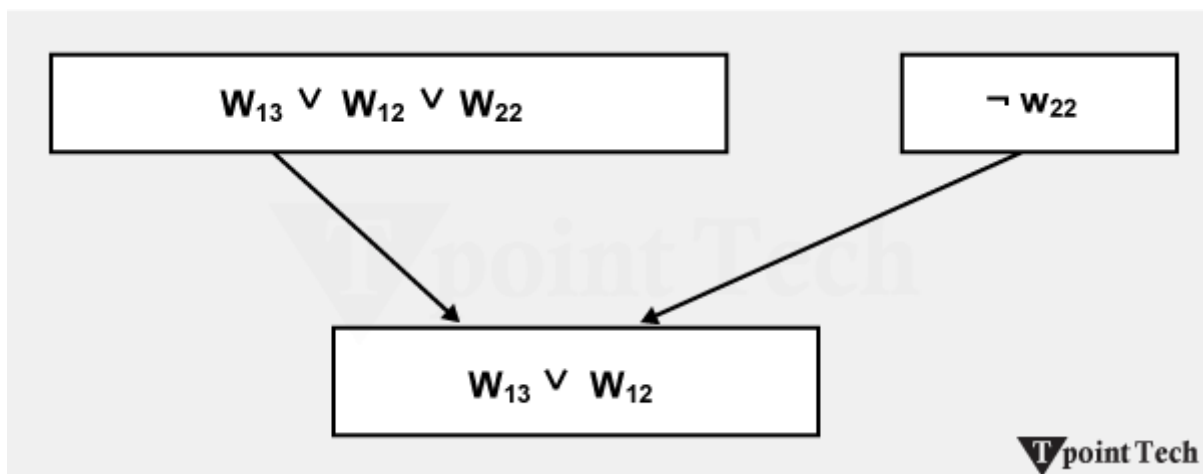
Apply Unit resolution on $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$ and $\neg W_{11}$:

After applying the Unit resolution formula on $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$ and $\neg W_{11}$ we will get $W_{13} \vee W_{12} \vee W_{22}$.



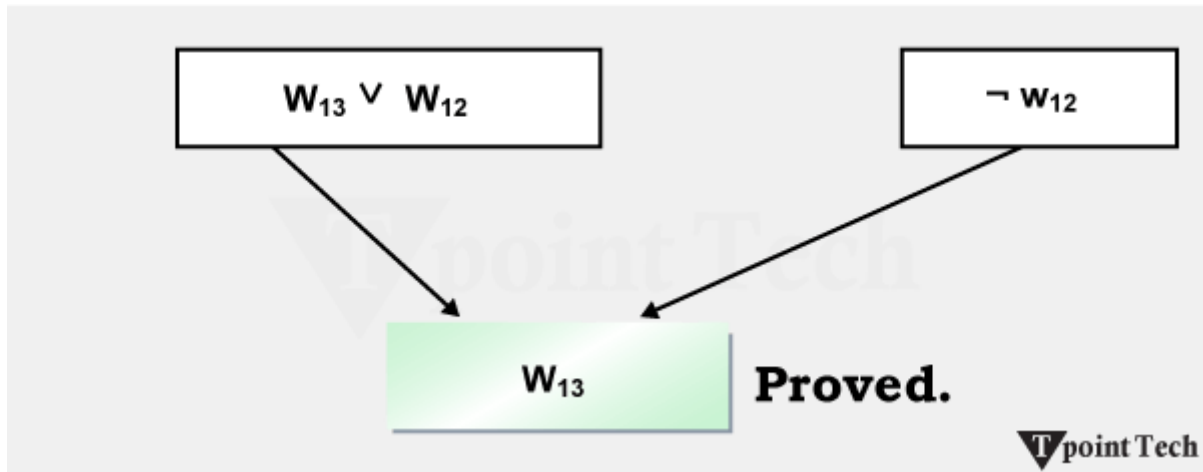
Apply Unit resolution on $W_{13} \vee W_{12} \vee W_{22}$ and $\neg W_{22}$:

After applying Unit resolution on $W_{13} \vee W_{12} \vee W_{22}$, and $\neg W_{22}$, we will get $W_{13} \vee W_{12}$ as output.



Apply Unit Resolution on $W_{13} \vee W_{12}$ and $\neg W_{12}$:

After Applying Unit resolution on $W_{13} \vee W_{12}$ and $\neg W_{12}$, we will get W_{13} as an output, hence it is proved that the Wumpus is in the room [1, 3].



Building the Knowledge Base for the Wumpus World

Encoding the Starting State of the Wumpus World (Easy Version)

1. Starting Setup

- The Wumpus World is a **4×4 grid** with rooms.
- The agent starts in the **bottom-left room**, which is Room [1,1].

Logical Statement:

$\text{At}(\text{Agent}, [1,1]) \rightarrow \text{The agent is at Room } [1,1].$

2. Unknown Danger at Start

- We don't know where the **Wumpus** or **pits** are yet.
- But we know there is **no pit** in the starting room.

Logical Statement:

$\neg \text{Pit}([1,1]) \rightarrow \text{No pit in Room } [1,1]$

3. Gold Location Unknown

- At the beginning, we also don't know where the **gold** is.

Logical Statement:

$\neg \text{Gold}([1,1]) \rightarrow \text{No gold in Room } [1,1]$

4. Safe and Unsafe Zones

- A room is **safe** if it has **no pit** and **no Wumpus**.

Logical Rule:

$\text{Safe}([x,y]) \Leftrightarrow \neg \text{Pit}([x,y]) \wedge \neg \text{Wumpus}([x,y])$

- If there is a **breeze**, it means a **pit is nearby**.

Logical Rule:

$\text{Breeze}([x,y]) \Rightarrow \text{Pit}([x+1,y]) \vee \text{Pit}([x,y+1])$

- If there is a **stench**, it means a **Wumpus is nearby**.

Logical Rule:

$\text{Stench}([x,y]) \Rightarrow \text{Wumpus}([x+1,y]) \vee \text{Wumpus}([x,y+1])$

5. Updating Knowledge with Percepts

- When the agent senses something (like stench or breeze), the knowledge base is updated.

Example:

If the agent smells a **stench** in Room [2,2], we add:

- $\text{Stench}([2,2])$
- $\text{Wumpus}([1,2]) \vee \text{Wumpus}([2,1]) \vee \text{Wumpus}([2,3]) \vee \text{Wumpus}([3,2])$

6. Marking Safe Rooms

- If the agent detects **no stench** and **no breeze** in Room [1,1], then nearby rooms are safe.

Logical Statement:

$\neg \text{Stench}([1,1]) \wedge \neg \text{Breeze}([1,1]) \Rightarrow \text{Safe}([1,2]) \wedge \text{Safe}([2,1])$

Handling Uncertainty in the Wumpus World

◆ What is Uncertainty?

Sometimes, the agent gets a clue (like a bad smell), but it's not clear **where the danger is**. This is called **uncertainty** — the agent has to **guess and explore** to find the truth.

◆ Example of Ambiguity

- The agent smells a **stench** in Room [2,2].
- That means the Wumpus could be in one of the nearby rooms:
→ [1,2] or [2,3] or [3,2]
- Logical form:
 $\text{Wumpus}([1,2]) \vee \text{Wumpus}([2,3]) \vee \text{Wumpus}([3,2])$
- The agent must **explore more rooms** to figure out the correct location.

◆ Logical Rules in the Knowledge Base

These rules help the agent make smart decisions:

Rule Type	Logical Statement	Meaning
Safe Movement	$\text{Safe}([x,y]) \Rightarrow \text{Move}(\text{Agent}, [x,y])$	If a room is safe, go there
Avoid Danger	$\neg \text{Safe}([x,y]) \Rightarrow \neg \text{Move}(\text{Agent}, [x,y])$	If a room is not safe, avoid it
Gold Detection	$\text{Glitter}([x,y]) \Rightarrow \text{Gold}([x,y])$	If glitter is seen, gold is there
Shoot Wumpus	$\text{Wumpus}([x,y]) \Rightarrow \text{Shoot}(\text{Agent}, [x,y])$	If Wumpus is found, shoot it

◆ Practical Example

1. **Agent smells stench in Room [2,1]**
 - Add: `Stench([2,1])`
 - Infer: Wumpus could be in [1,1], [2,2], or [3,1]
→ `Wumpus([1,1]) ∨ Wumpus([2,2]) ∨ Wumpus([3,1])`
2. **Agent finds no stench in Room [3,1]**
 - Add: `¬Stench([3,1])`
 - Infer: Wumpus is **not** in [2,1]
→ `¬Wumpus([2,1])`

Applications of the Wumpus World Knowledge Base

Wumpus World Knowledge Base – Easy Explanation

1. Using Logic

- Wumpus World uses **rules and facts** to describe the world.
- Example: “If there is a bad smell → Wumpus might be nearby.”

2. Finding New Facts

- AI can use methods like **forward chaining** and **backward chaining** to discover new things from old facts.

3. Handling Uncertainty

- Sometimes, the agent doesn't know everything.
- Wumpus World teaches how to **make decisions** even when some information is missing — like in **medical diagnosis** or **risk analysis**.

4. Solving Problems

- The agent uses smart search methods like **BFS**, **DFS**, or **A*** to find gold and avoid danger.

5. Using Heuristics

- Heuristics are smart guesses.
- The agent uses them to explore **safe rooms first**.

6. Step-by-Step Decisions

- The agent updates its plan as it learns new things — just like **robots** or **game AI**.

7. Understanding the Environment

- The agent uses clues like **smell**, **breeze**, and **glitter** to build a mental map — like **self-driving cars** do.

8. Making Its Own Decisions

- If the agent has enough knowledge, it can decide on its own — useful for **drones** or **industrial robots**.

9. Balancing Risk and Reward

- The agent learns to take smart risks — like **finance AI** or **healthcare bots** do when choosing between options.